Hongjia Liu

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Address: University of California San Diego, La Jolla, CA 92093, USA.

Education

Southern University of Science and Technology (SUSTech), Shenzhen, China *Sep. 2023 – present*

B.Eng. in Intelligent Medical Engineering, Department of Biomedical Engineering (expected in Jun. 2027)

GPA (overall): 3.87 / 4.00

Core courses: Digital Signal Processing (99), Big Medical Data (97), Intelligent Sensing Technology (95), Medical Robotics (95), Biomedical Optics (94), Medical Image Processing (92), Digital Circuits (92)

University of California San Diego (UCSD), La Jolla, CA, USA *Sep. 2026 – Dec. 2026*

University & Professional Studies (UPS) Program, Fall 2026 Exchange Program

Core courses: AI: Probabilistic Models, Deep Learning for Natural Language Understanding, Biosystems and Control

Publications & Patent

- **Hongjia Liu**, Wenjin Wang, *New Insights into the Influence of Illumination Geometry on Multi-Wavelength Photoplethysmographic Imaging,* manuscript submitted to ***IEEE Transactions on Biomedical Circuits and Systems*** *(Ref: TBioCAS-2026-Oct-0427-Reg).*
- **Hongjia Liu**, Huailei Lai, Huaijing Shu, and Wenjin Wang, *Contact vs. Non-Contact Lighting: Do We Understand What Remote-PPG is Measuring?*, **accepted** for the IEEE Engineering in Medicine and Biology Conference (**EMBC 2026**).
- Huaijing Shu, Huailei Lai, **Hongjia Liu**, and Wenjin Wang, *Camera-SpO_2 Measurement Using Time-Division Multiplexed Near-Infrared Imaging: A Feasibility Study on DC-based Calibration.* **Proceedings of EMBC 2025.**
- Wenjin Wang, Huaijing Shu, Huailei Lai, **Hongjia Liu**, *Blood Oxygen Saturation Detection Method, Device, Terminal Equipment, and Storage Medium Based on Pulsatile Signal Calibration*, Chinese National Invention Patent, **granted** (Patent No. ZL 202510679441.7).

Research Experience

Contactless Healthcare Lab, Southern University of Science and Technology *Sep. 2024 – Present*

Undergraduate Researcher (Advisor: Prof. Wenjin Wang)

Project I: Effects of Illumination Geometry on Multi-Wavelength PPG Imaging *Jul. 2025 – Present*

(IEEE TBioCAS, Independent First Author)

- Developed a dual-mode multi-wavelength remote photoplethysmography (rPPG) imaging system and signal processing framework to investigate contact and non-contact illumination effects in 37 participants.
- Revealed how illumination geometry affects **amplitude- and phase-domain rPPG signal formation** through RoR-based SpO_2 estimation and multi-wavelength pulse transit time (MW-PTT) analysis.

Project II: Spatial Analysis of rPPG Under Contact and Non-Contact Illumination *Mar. 2025 – Jan. 2026*

(IEEE EMBC 2026, Independent First Author)

- Developed a dual-mode rPPG imaging system and **spatial heatmap analysis** framework to investigate contact and non-contact illumination effects on rPPG signal formation, validated through 15 participants.
- Demonstrated that illumination geometry reshapes spatial distributions of AC, DC, AC/DC, and MW-PTT, revealing wavelength-dependent light-tissue interaction patterns.

Project III: SpO_2 Monitoring via Time-Division Multiplexed Near-Infrared Imaging *Sep. 2024 – Feb. 2025*

(IEEE EMBC 2025, Third Author)

- Contributed to a **time-division multiplexed near-infrared imaging** system for camera-based SpO_2 monitoring, demonstrating the potential of DC signals for multi-wavelength oxygenation estimation.
- Contributed to signal analysis and calibration for SpO_2 estimation in 11 participants, including investigation of temperature-induced camera baseline drift.

Selected Course Projects

Project I: EEG Emotion Recognition with ST-SHAP

Course Project — Machine Learning and its Medical Engineering Applications

- Adapted the **Swin Transformer-based ST-SHAP** framework to the EmoEEG-MC dataset for three- and seven-class EEG emotion recognition using spatial–temporal differential entropy features.
- Compared five frequency bands across video, imagery, and combined settings. Three-class experiments reached peak within-subject accuracy above **95%** and leave-one-subject-out accuracy exceeding **50%**.
- Applied **SHAP** to compare contributions from four consecutive EEG time segments, identifying the fourth segment as the strongest contributor to model predictions.

Project II: Active-Vision Camera Robot for Physiological Monitoring

Course Project — Medical Robotics

- Contributed to the design of an active-vision camera robot for frontal face tracking during treadmill exercise, supporting camera-based physiological monitoring.
- Developed **SolidWorks** models of the robotic assembly, established a **DH-based** forward kinematic model, and used MATLAB to simulate the robot's workspace and camera motion.

Project III: ECG Signal Processing and Arrhythmia Classification

Course Project — Digital Signal Processing

- Implemented a MATLAB pipeline for MIT-BIH ECG analysis, including IIR/FIR filtering, R-peak detection, ECG feature extraction, and SVM classification.
- Achieved **96.36%** accuracy in five-class classification with a mixed patient split and **79.80%** in three-class classification with a patient-disjoint split, evaluating performance on unseen patients.

Awards & Scholarships

Excellent Student Honor, SUSTech *2024, 2025, 2026*

Outstanding Student Scholarship, SUSTech *2024, 2025, 2026*

Futurui Scholarship, Department of Biomedical Engineering, SUSTech *2026*

Academic Excellence Award, SUSTech *2026*

Skills

Language: TOEFL iBT 5.5/6.0 (comparable score: 110/120).

Programming: MATLAB, Python, PyTorch, C/C++, Java

Biomedical Signal Processing: ECG, EEG, PPG, remote-PPG, Heart Rate, HRV, SpO_2, PTT, Blood Pressure

Machine Learning: Deep learning for physiological time series (e.g. CNNs, LSTMs, Transformers); classical machine learning (e.g. SVM, tree-based classifiers)

Modeling & Simulation: SolidWorks, Medical Robotics Modeling, MATLAB-based Kinematic Simulation